

IP BATTLES SHAPING INDIA'S EV AND MOBILITY SECTOR

Case Studies	Description	Takeaway
Mahindra Electric vs Counterfeit Battery Makers	Counterfeiters were selling cheap copies of Mahindra's modular EV battery packs. Mahindra demonstrated that its patent was being violated and obtained a temporary court order to halt the infringement.	The case reinforced the need for robust patent portfolios to combat counterfeits, especially given that EV components are easy targets for imitation.
Tata Motors vs Chinese Carmaker (Harrier Look-Alike)	A Chinese automaker mimicked the Tata Harrier's grille, headlamp structure, and body contours. The Delhi High Court ruled in Tata's favour, awarded ₹320 million in damages, and halted production.	Strong industrial design portfolios can decisively curb foreign design piracy.
Maruti Suzuki and the SmartPlay Infotainment Ecosystem	Maruti insured its infotainment system through copyright (software + UI) and design registration (dashboard layout), while cross-licensing with Google and Apple. This combination prevented reverse engineering of its UX.	Hybrid IP protection is becoming critical for connected-car UX.
Ola Electric vs Startup (Battery-Swapping Mechanism)	Ola secured a temporary injunction against a startup allegedly replicating its patented battery-swap mechanisms, showing how process patents can safeguard backend engineering, not just physical products.	Patents around EV infrastructure, especially swapping mechanisms, are emerging as major IP battlegrounds.

- *Incremental inventions.* Targeted engineering improvements, such as enhanced cooling geometries, safer cabin configurations, more efficient switching sequences, or refined drivetrain assemblies

- *Breakthrough inventions.* New propulsion concepts, advanced EV architectures, transformative safety mechanisms, and other disruptive leaps

Both qualify for protection equally if they are novel, useful, and not obvious to an experienced engineer. Simple, smart fixes that solve specific technical problems usually make the most valuable automotive patents, often more so than even major advanced inventions.

Industrial designs. This protection guards a product's visual identity, covering UI layouts, dashboard designs, exterior styling, packaging, and overall form language. As cockpits become screen-centric and UX takes centre stage, automobile manufacturers are increasingly filing design patents. While design rights protect only visible elements, they can cover distinctive shapes, contours, lighting geometries, and interface visuals, though hidden structures remain unprotected.

Copyright. Software source code, UI artwork, icons, human-machine interaction elements, and infotainment layouts are protected

by copyright. While India does not allow patents on pure software algorithms or business methods, the original expression of the code and the visual structure of its interface can be copyrighted. This protection is increasingly vital as reverse-engineering of UX layouts and software behaviour becomes easier.

Trademarks. They secure brand assets, such as product names, interface labels, emblems, audio signatures, colour palettes, and distinctive forms. In modern automotive branding, this frequently expands to sensory signals, engine-ignition sounds, lighting sequences, loading visuals, and user-interaction patterns that strengthen subconscious brand recall.

Lesson in incremental vs breakthrough innovation

Patentability in the automotive world becomes clearer when viewed through the long evolution of mobility itself. The open bullock carts I saw earlier evolved into enclosed compartments, then into cabins that protected against the weather, and finally into the aerodynamic shapes of today's vehicles. Across each stage of this evolution, I have observed incremental design and engineering changes aimed at improving safety, comfort, efficiency, or durability.

In contrast, the industry has also witnessed significant innovations,

with the shift dramatic. Steam propulsion gave way to internal combustion, and later combustion systems transitioned to electric powertrains. These represent the technological leaps that redefine the fundamentals of mobility. Each transformation is as patentable as incremental improvements, demonstrating how both evolutionary and revolutionary innovations contribute to the expanding IP landscape in automotive engineering.

As an investor, I follow a simple principle: I first identify a clearly defined problem and then determine whether a unique solution exists. Once both elements are evident, filing a patent becomes a logical next step.

Data: The new motor running inside every vehicle

Connected vehicles today generate immense volumes of data, ranging from location intelligence and driving behaviour to ADAS outputs, telematics logs, lidar perception, human-machine interaction patterns, and predictive insights. As vehicle data becomes integral to performance, safety, and personalisation, OEMs are exploring new ways to derive value from it.

This momentum has triggered a wave of patents in connected-vehicle analytics, data-to-wearable communication, behavioural intelligence engines, predictive maintenance